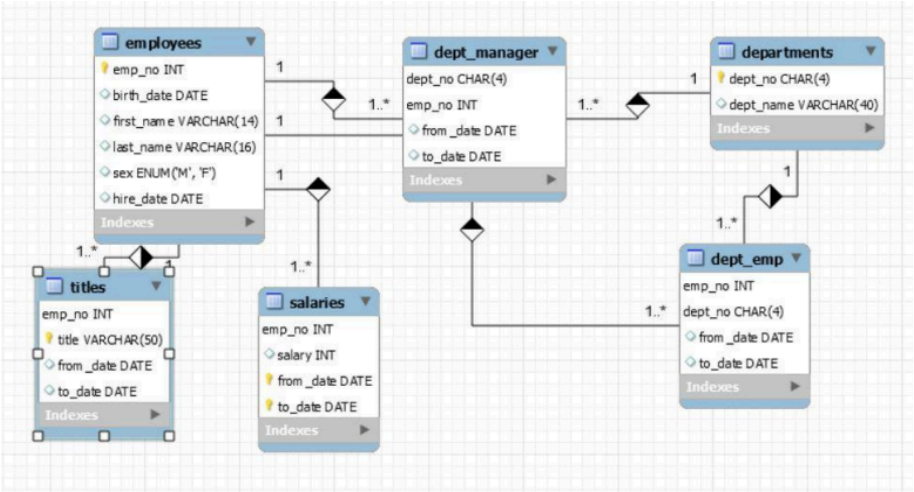
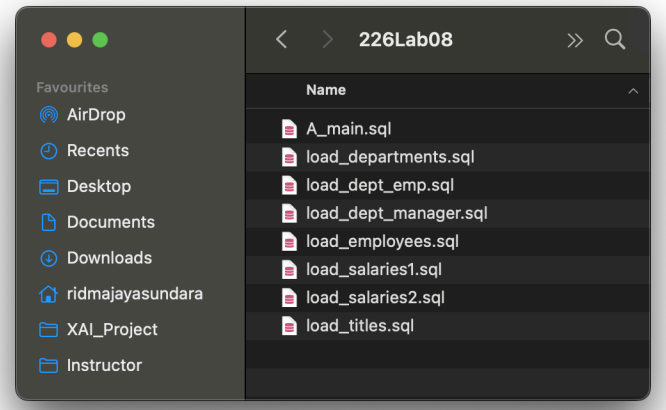
CO226-Apr2025 : Database Systems

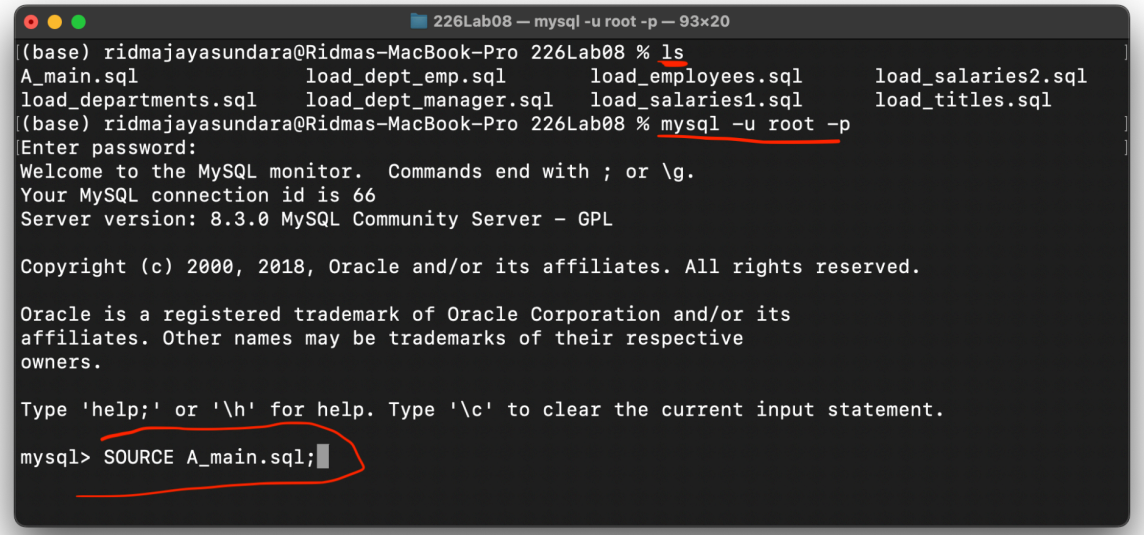
Lab 08 : Pre Lab

Refer to the Company ER diagram shown in Figure below.



.sql scripts to create the database, create the table and populate are given in the pre lab folder. Open the terminal inside the lab folder, log in to mysql and use the source “A_main.sql”.





This will create the database named E21Company, and create all the tables that are mentioned in the ER diagram and then populate all the tables.

Feel free to go through A_main.sql and other .sql files to see what has been done. Now that the database setup is done let's move on to the tasks.

Pre Lab : Part 1

Save each query from tasks below to a .sql file, you would need to use these again in the lab.

1. List the number of Engineers each department has
2. List all the female employees who are department managers and have worked as a senior engineer
3. Display the departments and titles of employees who have a salary greater than 115000. Display how many of such employees work for each department.
4. Assume that the company wants to reward the most senior employees who are more than 50 years of age and have contributed to the company for more than 10 years. Who is on the list? Display employee name, age, years of service in the company and joined date
5. Find all the names (first name + last name) of employees in the database who do not work in the Human Resources department. Assume that all the people work for exactly one department

Pre Lab : Part 2

Go through perusall AND these below references,

- https://dev.mysql.com/doc/refman/8.0/en/mysql-indexes.html
- https://www.w3schools.com/sql/sql_create_index.asp

- https://dev.mysql.com/doc/refman/5.7/en/using-explain.html

And prepare a 1-2 page **handwritten** summary on Indexing and Query Optimization.

Prepare your prelab work into a single pdf document named E21XXX_CO226_Lab08_PreLab.pdf

Pages 1 and 2 of the pdf must contain your handwritten summary for Part 2 of the prelab, covering Indexing and Query Optimization based on the provided resources. The remaining pages should include screenshots of the SQL queries and outputs from Part 1.

Submit it no later than 8:00 AM on Tuesday, 8th July.